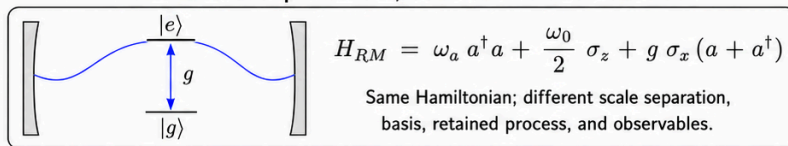


One microscopic model, several effective theories



near resonance

far detuned

displaced basis

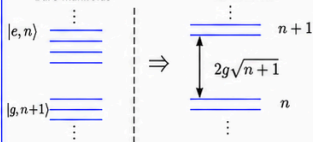
singular scaling

Near resonance / weak coupling

$$\frac{g\sqrt{n_*+1}}{\Sigma} \ll 1 \quad \frac{|\Delta|}{\Sigma} \ll 1$$

Bare manifolds

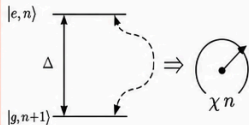
JC doublets



- Natural basis:** Bare excitation manifolds
- Retained process:** Slow one-excitation exchange
- Effective theory:** RWA / Jaynes-Cummings
- Physical content:** Rabi oscillations, dressed doublets, collapse and revival
- Failure:** Counter-rotating error depends on state, occupation, observable, and time

Far detuned

$$\frac{g\sqrt{n_*+1}}{|\Delta|} \ll 1$$

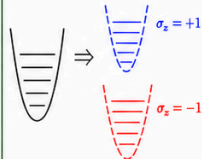


- Natural basis:** Bare states with virtual admixture
- Eliminated process:** Real excitation exchange
- Effective theory:** Dispersive Jaynes-Cummings
- Key scale:** $\chi = g^2/\Delta$
- Physical content:** Conditional shifts, QND structure, parity mapping, virtual bus, Purcell decay
- Failure:** $n \gtrsim n_{\text{crit}}$ or long-time phase accumulation $n_{\text{crit}} = \frac{\Delta^2}{4g^2}$

Intermediate to deep strong coupling

1. Polaron transformation / GRWA
2. Deep-strong displaced oscillators

$$U_p = \exp[-(g/\omega_a)\sigma_z(a^\dagger - a)]$$



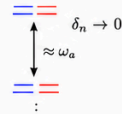
Retain selected couplings between displaced manifolds

$$\frac{|\text{discarded matrix element}|}{|\text{energy gap}|} \ll 1$$

Failure: Incorrect displaced-manifold assignment or uncontrolled inter-manifold coupling

$$b_s = a + s \frac{g}{\omega_a}$$

$$H_{\text{DSC}}^{(0)} = \omega_a b_s^\dagger b_s - \frac{g^2}{\omega_a}$$



Harmonic return, parity doublets, conditional trajectories, collapse and revival

$$n_{\text{bare}} = (a^\dagger a) \quad n_{\text{disp}} = (b^\dagger b) \quad n_{\text{det}} = (X^- X^+)$$

Bare occupation $\neq$ displaced excitation $\neq$ detector count

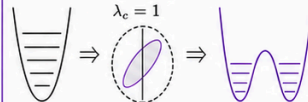
Large-frequency-ratio limit

$$\frac{\omega_0}{\omega_a} \rightarrow \infty$$

at fixed

$$\lambda = \frac{2g}{\sqrt{\omega_0 \omega_a}}$$

$$\lambda_c = 1$$



Normal phase:

$$\epsilon_{\text{NP}} = \omega_a \sqrt{1 - \lambda^2}$$

Superradiant phase:

$$\epsilon_{\text{SP}} = \omega_a \sqrt{1 - \lambda^{-4}}$$

- Physical content:** Squeezing, soft-mode gap closing, two displaced minima, parity doublets
- Important distinction:** Finite-ratio crossover $\neq$ singular quantum phase transition
- Failure:** Finite-frequency rounding or insufficient oscillator cutoff

These theories are not a hierarchy of universal accuracy.

Each one retains a different physical structure.